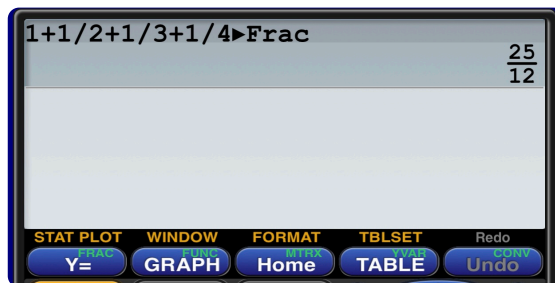


Section 5.1 Riemann Sums

Notation $\sum$ is used for "sum" in math

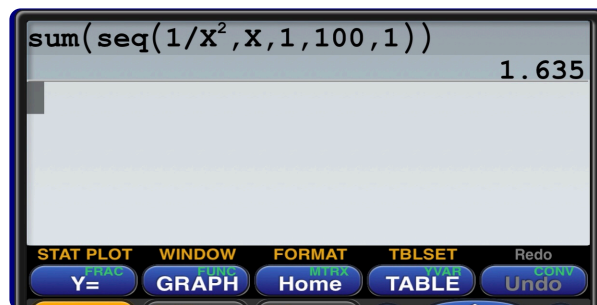
Ex ① $\sum_{k=1}^4 \frac{1}{k} = \frac{1}{1} + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} = \frac{25}{12}$



What if there are 100 terms to add up?

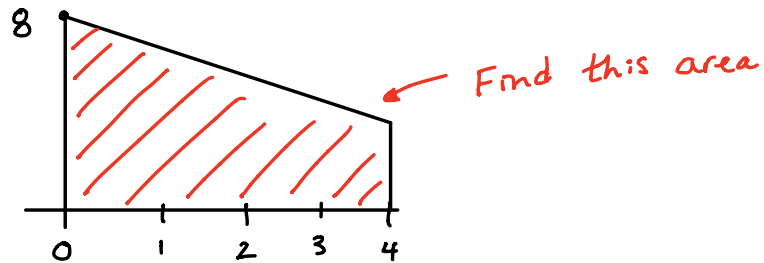
Use $\text{sum}(\text{seq}(\text{fnc}, x, \text{start}, \text{stop}, \text{increment}))$
on the graphing calculator.

Ex ② $\sum_{k=1}^{100} \frac{1}{k^2} = \text{sum}(\text{seq}(\frac{1}{x^2}, x, 1, 100, 1)) = 1.635$

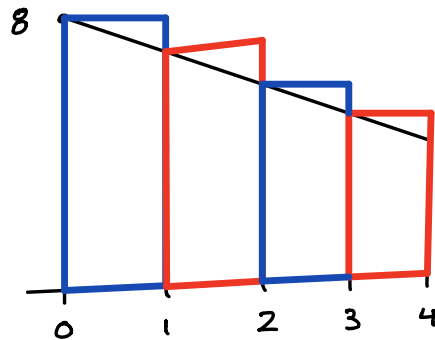


Ex ③ $f(x) = -x + 8$ on $[0, 4]$

Find the area between $f(x)$ and x -axis.



We will use 4 rectangles to estimate the area.

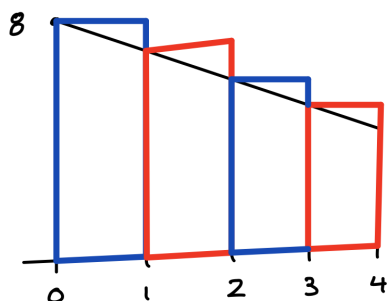


The width of each rectangle is 1.

The height is found by substituting the left hand endpoint of each rectangle into $f(x)$.

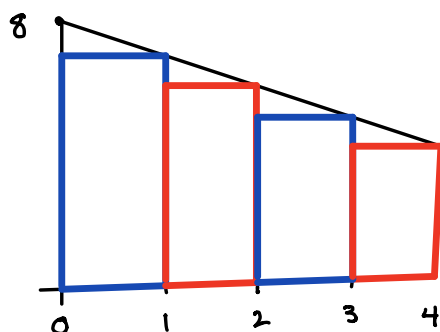
$$A \approx (-0+8) \cdot 1 + (-1+8) \cdot 1 + (-2+8) \cdot 1 + (-3+8) \cdot 1$$

$$A \approx 8 + 7 + 6 + 5 = \textcircled{26}$$



The area of these 4 rectangles is an overestimate for the area between $f(x)$ and the x -axis.

Next, use the right hand endpoints to find the height of each rectangle.

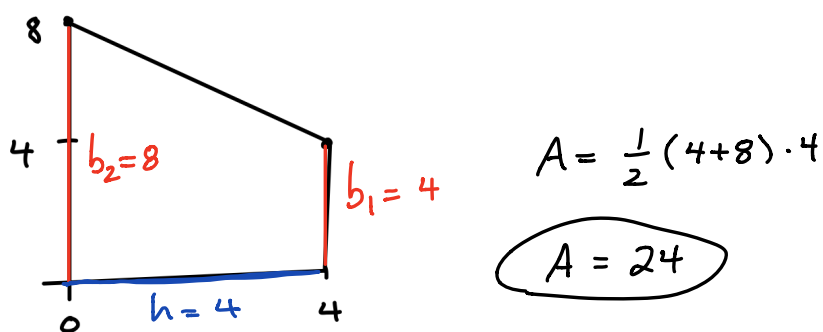
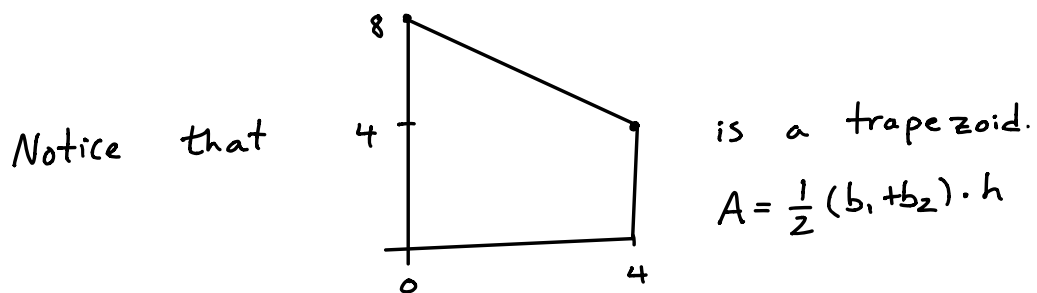


$$\text{Area} \approx [-1+8] \cdot 1 + [-2+8] \cdot 1 + [-3+8] \cdot 1 + [-4+8] \cdot 1$$

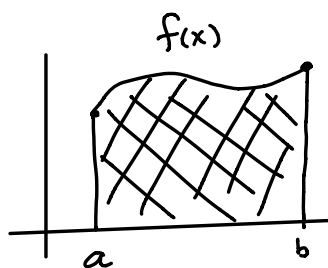
$$\text{Area} \approx 7 + 6 + 5 + 4 = \textcircled{22}$$

Notice that this is an under estimate for the area. If we average the two answers, we get an even better estimate.

$$A \approx \frac{22+26}{2} = \textcircled{24}$$

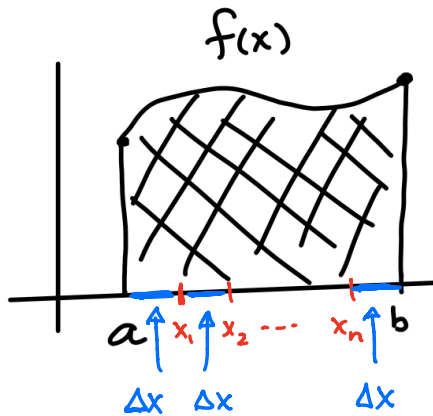


To find area between $f(x)$ and x -axis on $[a, b]$



Use n rectangles to estimate the area.

$\Delta x = \frac{b-a}{n}$ is the base of each rectangle



For each subinterval, pick x_i .

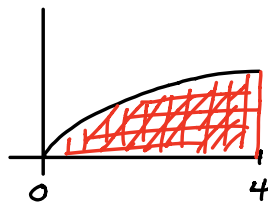
$$\text{Area} \approx \sum_{i=1}^n \underbrace{f(x_i)}_{\text{height}} \cdot \underbrace{\Delta x}_{\text{base}}$$

As n gets larger, the estimate will be better.

So, let $n \rightarrow \infty$ for the actual Area.

$$\text{Area} = \lim_{n \rightarrow \infty} \left(\sum_{i=1}^n f(x_i) \cdot \Delta x \right)$$

Ex ④ Consider the area between $f(x) = \sqrt{x}$ and the x -axis on $[0, 4]$

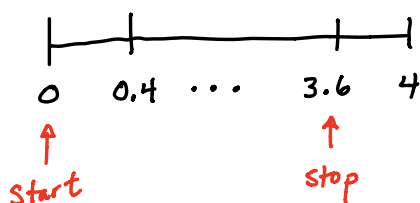


Estimate the area using n rectangles.

A) left hand endpoints of $n = 10$ rectangles

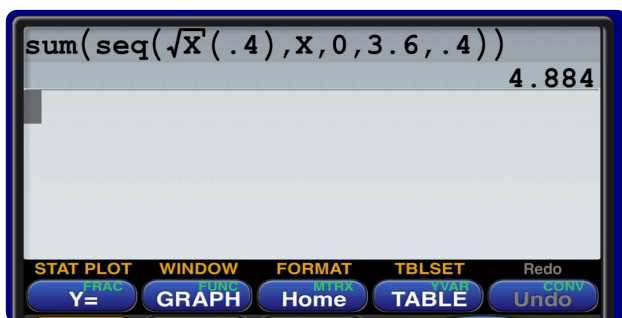
$$\Delta x = \frac{4-0}{10}$$

$$\Delta x = 0.4$$



$$\text{sum}(\text{seq}(\sqrt{x}(0.4), x, 0, 3.6, 0.4))$$

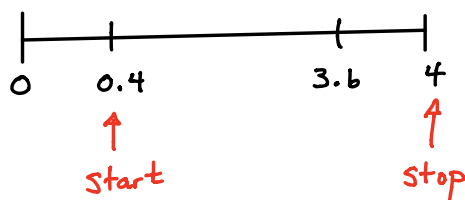
put Δx in both places



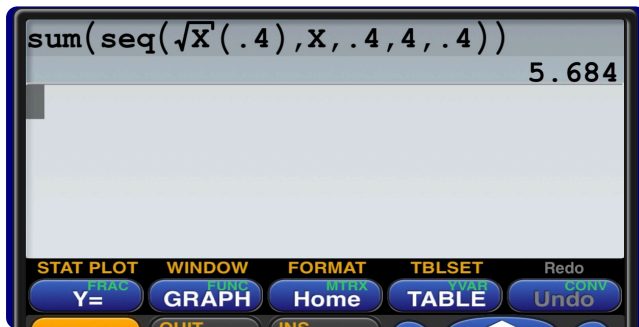
$$A \approx 4.884$$

B) right hand endpoints of $n = 10$ rectangles

$$\Delta x = 0.4$$

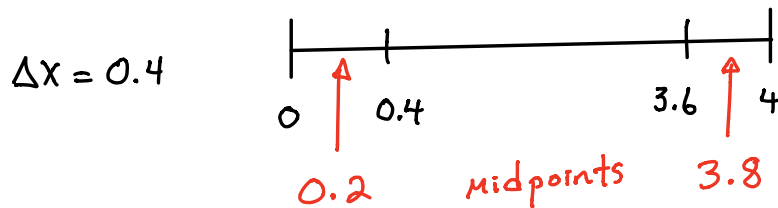


$$\text{Sum}(\text{seq}(\sqrt{x}(.4), x, 0.4, 4, 0.4))$$

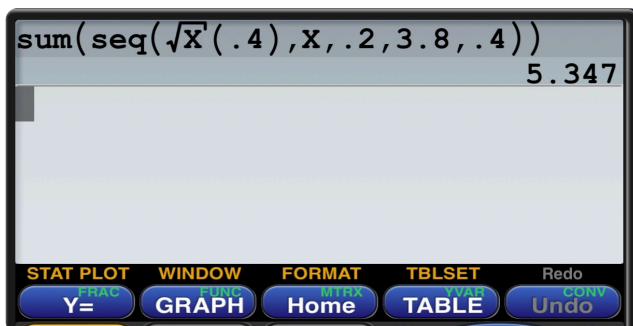


$$A \approx 5.684$$

c) use midpoints of $n = 10$ rectangles.



$$\text{Sum}(\text{seq}(\sqrt{x}(.4), x, 0.2, 3.8, 0.4))$$



$$A \approx 5.347$$

Now use the calculator to find the
Actual area.

